Learning Nonprehensile Manipulation with A Snake Robot

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Abstract

Snake robots offer unique advantages in the exploration of unstructured domains due to their highly adaptable morphology. While historical snake robot research has primarily surrounded locomotion, the form factor possesses strong potential as a mobile manipulator in extreme environments. However, the control of snake robots for manipulation tasks remains a challenge. Traditional end-effector-based methods often underutilize the robot's anatomy, while classical whole-body controllers struggle to manage the high-dimensional state space and complex contact modeling. In this work, we propose a reinforcement learning framework for autonomous, nonprehensile manipulation using the Crater Observing Bio-inspired Rolling Articulator (COBRA) system, a modular snake robot. We train a control policy in a high-fidelity physics simulator to perform a planar pushing task. Our approach uses a single keypoint-based reward for error tracking and leverages curriculum learning to efficiently guide the policy towards robust whole-body behaviors. Simulation results demonstrate that the learned policy smoothly coordinates the robot's 11 joints to translate and rotate a target object to a goal pose. The policy achieves a near-perfect success rate on simulated rollouts. This project demonstrates the effectiveness of reinforcement learning in achieving natural manipulation behaviors with a snake robot. The results provide a foundation for future work in sim-to-real transfer and locomanipulation in the field.

Introduction

Snake robots are highly adaptable robotic platforms capable of operating in diverse and unstructured environments. This adaptive ability has been demonstrated in the field, where snake robots have been used to investigate arctic caves (Vaquero et al. 2024), assist urban search and rescue operations (Whitman et al. 2018), and climb sandy slopes (Marvi et al. 2014). The redundant morphology of snake robots enables them to utilize a variety of locomotion modes depending on the environment and desired motion. They can sidewind across large distances (Tesch et al. 2009), roll up tree trunks (Enner, Rollinson, and Choset 2013), and crawl through narrow pipes (Inazawa et al. 2020). Some designs are specialized even further for the mission specifications, such as including a unique head and tail module that mechanically lock together to enable tumbling down crater slopes in a wheel-like formation (Salagame et al. 2026). This multimodal ideology extends to the domain of mobile manipulation. Typical mobile manipulation platforms have a clear distinction between locomotion and manipulation in their design, often realized in the form of a robot arm mounted on a moving base (Fu, Zhao, and Finn 2024) (Sleiman, Farshidian, and Hutter 2023). Snake robots do not fall into this category. Instead, the line between locomotion and manipulation is blurred, as each part of the body can be leveraged to both move the robot and manipulate objects in the environment.

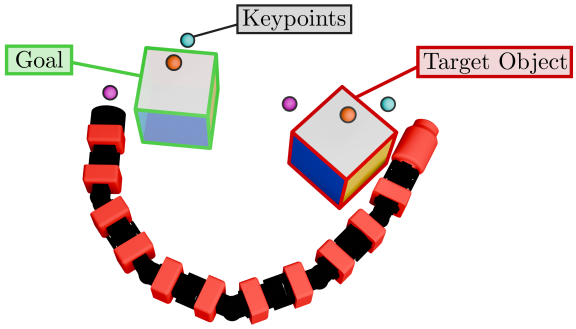


Figure 1: COBRA performing nonprehensile manipulation to push a box to a goal configuration.

The main control problem for mobile manipulation with arm-style platforms is to intelligently fuse the locomotion and manipulation control stacks, e.g., bending the front legs to allow the manipulator to reach an object on the ground. Since snake robot platforms do not contain a structural separation of concerns, the control problem is much more open-ended. In this work, we aim to address the snake robot manipulation problem by using reinforcement learning to train an autonomous box-pushing policy. The snake robot is simulated in a state-of-the-art robotics simulator and given a basic planar pushing task. As the snake robot interacts with the environment, the reinforcement learning agent explores the high-dimensional state space and incrementally adjusts the control policy to maximize expected return. Through this exploration, the control policy can converge to diverse and natural whole-body behaviors that would be difficult to discover classically. The reward structure is carefully shaped to

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encourage smooth and robust motions and the performance of the final control policy is evaluated in simulation.

Related Work

Existing research in snake robot manipulation can be divided into two categories: end-effector-based (EE) control and whole-body (WB) control. EE methods refer to architectures which explicitly designate a group of links to form a stable base, while the remaining links act as a standard robotic arm. In these methods, the major algorithmic challenge is to establish a stable configuration that allows the arm links to act freely without tipping over the body or otherwise impeding the task. This style of control is more prevalent in the literature as once a stable base is established, the core manipulation capabilities can leverage the rich history of manipulator control. Ji et al. adopt this style to demonstrate semi-autonomous grasping with a dedicated 1-DOF gripper module on a small-scale snake robot (Ji et al. 2023). Elsayed et al. showcase EE control in a different domain, as they accomplish manipulation tasks with a snake robot while navigating a horizontal pipe with a helical gait (Elsayed et al. 2022). The decoupled nature of EE control can be effective in basic manipulation tasks, however, it often limits the workspace of the robot and does not fully take advantage of the snake robot anatomy.

The WB class of solutions allow for any part of the body to contribute to the manipulation task. Typically, this leads to nonprehensile behaviors, meaning the robot manipulates the target object without explicitly grasping with a gripper—as seen in the EE methods. Under this formulation, any point along the body can make and break contact with the target object, meaning many of the methods leveraged in basic manipulator control break down due to the curse of dimensionality. WB control for snake robot manipulation is a sparse area of research. To find related works, we can look to both the fixed-base manipulator domain and examples of mobile manipulation with more standard platforms. Levé et al. demonstrate whole-body manipulation onboard a dual-arm humanoid robot, using contact-implicit optimization to jointly solve for the state trajectory and contact points (Levé et al. 2025). However, this approach occurs at the kinematic level and is not feasible for real-time control even with the simplifications introduced in the modeling. In contrast, Jeon et al. propose a learning-based system to enable locomanipulation of diverse objects with a quadruped robot (Jeon et al. 2024). Instead of attempting to optimize over the infinite contact modes, the team from KAIST constructs the problem in a robotics simulator and leverages hierarchical reinforcement learning to train policies that run in real-time and can adapt to changing environmental conditions. In our work, we adapt the latter approach as classical methods can struggle with the highly redundant morphology and large number of contact modes that are characteristic of snake robot interaction.

Background

In this project, we perform manipulation using the Crater Observing Bio-inspired Rolling Articulator (COBRA) platform. COBRA is a snake robot consisting of 11 identical joint modules connected in a serial chain in an alternating fashion, yielding six pitching joints and five yawing joints. Figure 2 briefly introduces the modeling of the system. We assign a coordinate frame to each link on the robot and construct the virtual chassis (Rollinson, Buchan, and Choset 2012) at the center of mass of the robot. This frame is denoted by b to represent the base frame. In a full $\mathbb{R}^3$ formulation, the orientation of the virtual chassis is determined by SVD, which finds the principal axes that best describe the current shape of the robot. For the planar case, we simplify this rotation calculation by aligning the Z axis of the virtual chassis with the world frame and computing the yaw angle as an average of the yawing joints.

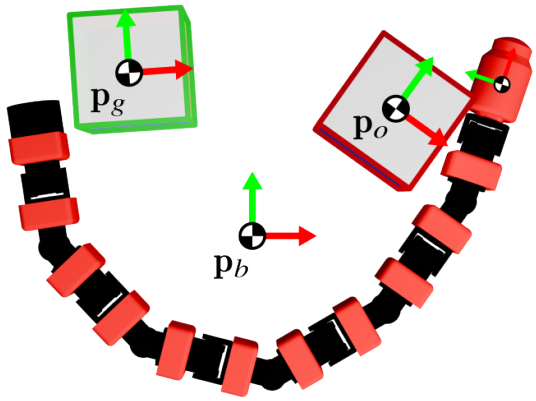


Figure 2: Top-down view of annotated coordinate frames for the box-pushing task.

To train the control policy we use Proximal Policy Optimization (PPO) (Schulman et al. 2017) for its stability and pervasiveness in robot learning literature. Figure 3 displays a block-style visualization of the network architecture used in the reinforcement learning formulation. PPO is an actor-critic architecture, where π_θ and V_ϕ denote the actor and critic network, respectively. The key characteristics of PPO lie in its update behavior, as it introduces conservative rules to prevent significant performance collapse. In future work, this architecture could be extended to provide extra privileged information to the critic network.

Project Description

To evaluate nonprehensile manipulation with a snake robot we choose a planar pushing task. The problem is formulated such that the task can be trivially extended to the non-planar case in future work. In the planar task, COBRA is instructed to push a $20cm$-sided cube to a goal pose. At the beginning of an episode, the initial pose and goal pose of the box are randomly sampled in a nearby neighborhood in front of the robot. The position is sampled in the XY plane and the pose is sampled as a heading angle. The task is considered complete when the box maintains a settled state for over N_s consecutive policy steps. A settled state is achieved when the position error, orientation error, linear velocity, and angular

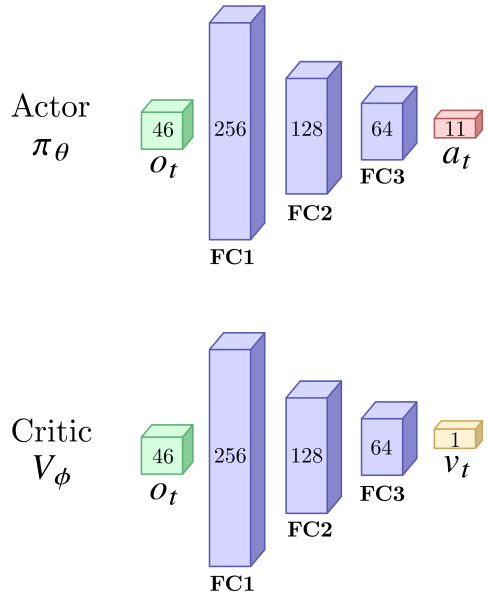


Figure 3: Actor-critic architecture utilized to train the control policy.

velocity are below specified thresholds. This ensures that the box is in a stable state that is sufficiently similar to the goal pose. If this condition is not achieved within a specified time limit, the episode terminates and the process repeats.

We frame the planar pushing task as a Markov Decision Process (MDP) with the following formulation:

Observation Space: The observation space consists of state measurements of the robot, box, and goal. The first terms are the relative joint positions and velocities, denoted by $\mathbf{q}, \dot{\mathbf{q}} \in \mathbb{R}^{11}$. Since the default configuration of COBRA is the zero position for all joints, this simply resolves to the nominal joint positions and velocities. The pose of the object and the goal are included as $[{}^b\mathbf{p}_o, {}^b\mathbf{R}_o] \in \mathbb{R}^{12}$ and $[{}^b\mathbf{p}_g, {}^b\mathbf{R}_g] \in \mathbb{R}^{12}$, where $\mathbf{R} \in \mathbb{R}^9$ represents a flattened rotation matrix in SO(3). These poses are represented with respect to the virtual chassis frame, which is denoted by b in Figure 2 and serves as the stable base frame of COBRA.

$$O = [\mathbf{q}, \dot{\mathbf{q}}, {}^b\mathbf{p}_o, {}^b\mathbf{R}_o, {}^b\mathbf{p}_g, {}^b\mathbf{R}_g]$$

Action Space: The action space is configured as joint commands relative to the current configuration of the robot. The actions are sampled from $[-1, 1]$ radians and scaled by 0.2. These are used as goal position commands for the internal PD actuator model in Isaac Lab.

$$\mathcal{A} = [\Delta\mathbf{q}], \quad \mathbf{q}_{cmd} = \mathbf{q}_{curr} + 0.2\,\mathbf{a}$$

Rewards: Table 1 lists the five reward terms and their respective weights in the final reward function. The main tracking reward is a measure of the error between the current pose of the object and the goal pose of the object. Inheriting from the work by (Dadiotis et al. 2025), we adopt a keypoint-based tracking reward. Instead of constructing separate rewards for both position error and orientation error, we use an average Euclidean distance between three predefined keypoints offset from the center of mass of the object. A visualization of these keypoints can be seen in Figure 1. As shown in the literature, the continuous nature of this tracking reward results in more stable training and better performance in learning the object manipulation task (Allshire et al. 2022). The other notable reward is the settled bonus reward that the agent receives for keeping the box in a settled state. Since a successful termination is only triggered if the box remains settled under the target thresholds for N_s steps, this settled bonus reward encourages the agent to let the box rest in the settled position in the early stages of learning. This was introduced to densify the task completion reward to speed up training. Without this reward, the agent is less likely to wait for the box to satisfy the settled condition. The other rewards follow the standard formulation for an object manipulation task, and consist of a time penalty, torque penalty, and task completion bonus.

Reward Term	**Expression**	**Weight**
Keypoint Tracking	$\sum_i^{n_k} \|k_{i,p} - k_{i,g}\|$	-10.0
Settled Bonus	$\mathbb{1}_{n_s>0}$	5.0
Task Completion	$\mathbb{1}_{n_s \geq N_s}$	100.0
Time Penalty	1	-0.5
Torque Penalty	$\|\tau\|^2$	-0.005

Table 1: Summary of reward terms and their associated weights.

Curriculum: There are two success-based curricula implemented to shape the training dynamics. The first is a gradually increasing difficulty on the thresholds that define a settled state. The position error and orientation error are decreased from 0.2m and 0.8rad to 0.05m and 0.3rad, respectively. This was implemented to prioritize learning a general manipulation strategy early on, rather than satisfying strict termination conditions. The second curriculum is on the settled bonus weight. This is gradually decreased from 5 to 0 as the agent begins to succeed in the task. Without removing this bonus, the agent could potentially learn to exploit the incremental reward instead of prioritizing the task completion reward. Both curricula are guided by an exponential moving average (EMA) of episode success rate.

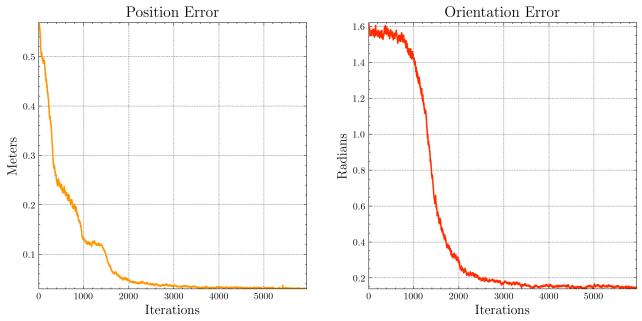
Experiments

The reinforcement learning formulation was implemented in NVIDIA's Isaac Lab (Mittal et al. 2025). A Universal Scene Description (USD) model of the COBRA robot was constructed and imported along with a custom cube asset. All experiments were run on a single RTX 3090 GPU, with 4096 environments simulated in parallel. The policy was trained at a 50Hz control rate for 5000 iterations, with 24 policy steps and 96 physics steps per iteration. The time step for physics integration was set to 0.005s (200Hz).

Figure 4 displays the metrics from the final training run. In Figure 4c, we see a smooth decrease in the tracking error on the keypoints, indicating the policy is learning to align the box with the goal pose. The position error and orientation error from Figures 4a and 4b tell the same story, converging

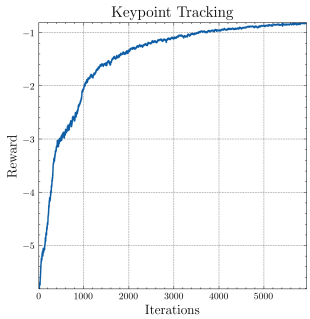
to values that closely correspond with the thresholds defined in the termination condition.

Figures 4d–4f demonstrate the influence of curriculum learning in the training. The dashed pink line at the 1400th iteration indicates the point at which the episode success rate triggers a tightening of the error thresholds and a weakening of the settled bonus weight. The effect of this event is clearly seen in Figure 4d as the performance of the policy dips slightly before continuing to improve.

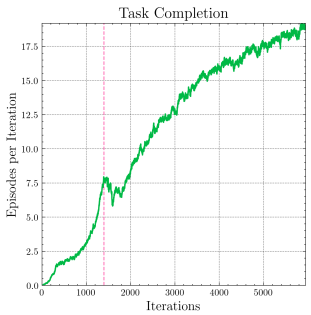


(a) Position Error

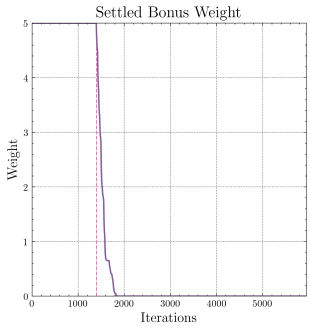
(b) Orientation Error



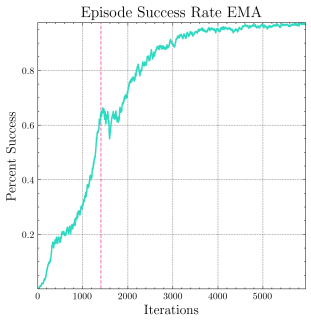
(c) Keypoint Tracking



(d) Task Completion



(e) Settled Bonus Weight



(f) Episode Success Rate EMA

Figure 4: Metrics from the final policy training. The pink dashed line marks a transition in the curriculum difficulty.

A significant fraction of the project effort was spent in the reward shaping phase, progressively adjusting the set of terms to guide the agent to complete the box-pushing task. The two most influential adjustments were the inclusion of the settled bonus curriculum and the keypoint tracking formulation. These changes resulted in a significant increase in the task completion performance during training. Figure 5 displays snapshots of the final policy in a rollout. The policy efficiently leverages the entire body of COBRA to perform nonprehensile manipulation of the box and push it to the goal pose in four seconds.

Conclusion

We present a reinforcement learning formulation for nonprehensile manipulation with a modular snake robot. A planar pushing task is selected to demonstrate complex multi-contact reorientation behaviors. We simulate the environment in a modern robotics simulator and leverage a state-of-the-art reinforcement learning algorithm to train a policy for joint control. Inheriting from the literature, the reward structure is shaped to enable efficient exploration. The policy is trained in a few hours on a single desktop GPU. In simulation, the policy demonstrates smooth, natural manipulation behaviors that achieve a high success rate in completing the box-pushing task.

Given the results from this project, the natural next step is to attempt to transfer the control policy to the real world system. Bridging the sim-to-real gap in this context will require significant effort, both in the learning dynamics and the simulation modeling. To begin, the actuator model in simulation needs to be adjusted to better match the actuator behavior on the real platform. This entails substituting the existing implicit PD actuator with a learned actuator network or an explicit delayed PD actuator with parameters determined from system identification. On the learning side, domain randomization could be introduced to attempt to increase the likelihood that the real world dynamics fall ”in-distribution” of the trained model. These steps will bring the policy closer to deployment in the real world, enabling robust snake robot manipulation.

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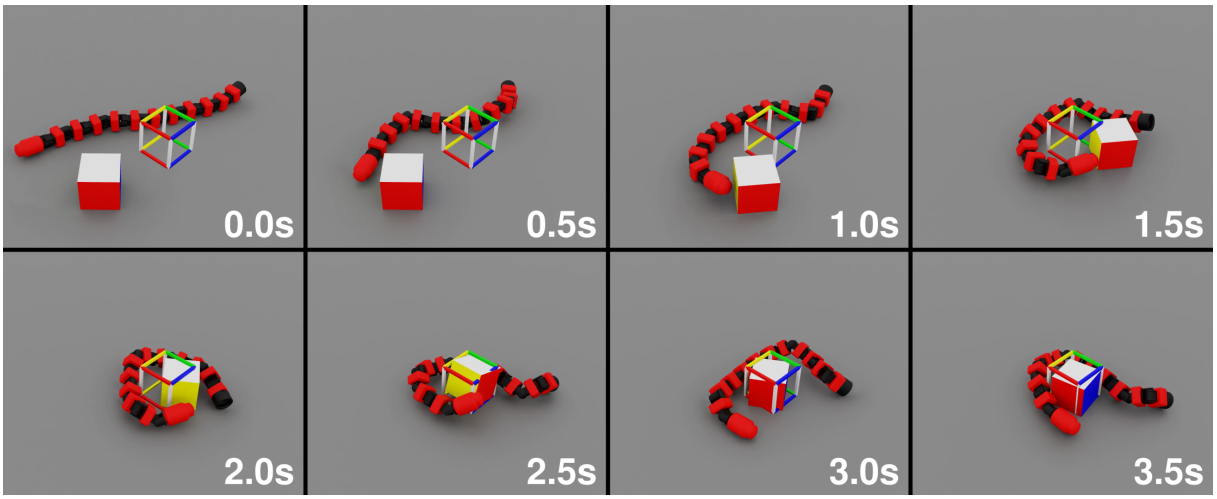


Figure 5: Snapshots of a policy rollout in Isaac Lab.

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